SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I 192126026 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks
Concept Identification	20%	2	2	2
Concept Mapping Structure	25%	2	2	2
Linking & Relationships	20%	1	1	1
Application & Analysis	20%	1	1	1
Explanation & Clarity	15%	1	1	1

Total Marks 21 /30

Signature of the Faculty

CONCEPT MAPPING QUESTIONS

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.
Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3.Given the concept map:

DataLink→MACAddress→LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→DataLinkLayer→ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✔ → Data Link ✔ → Network ✖ → Transport ✖ → Application ✖

Analyze the problem and explain how the concept map helps in troubleshooting.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships

CO1 AT - 2 Concept Mapping

Name : Ajay Pap

code : CSA 0710

course subject : computer networks

Serial no: 04

Questions :

(1) A partially completed concept mapping is given below:

Application → Presentation → _____ → Transport →

network → _____ → Physical Analyze the missing layer and explain how their function are essential

(2) Construct a concept map linking data unit (segment, packet, Frame, bits) with their respective OSI layer.

(3) Given the concept map:

Data link → MAC address → Local Delivery

Network → IP address → Global delievery

explain how these relationship ensure complete data.

7. Application Layer (user service & network Application)

↓

6) Presentation Layer (Encryption, compression, Formatting)

↓

5. Session Layer (Establishes and manages session)

↓

4. Transport Layer (Reliable Delivery and error)

↓

3. Network layer (IP addressing and routing)

↓

2. Data link layer (Framing, Mac address)

↓

1. Physical Layer (Transmit bits through physical media)

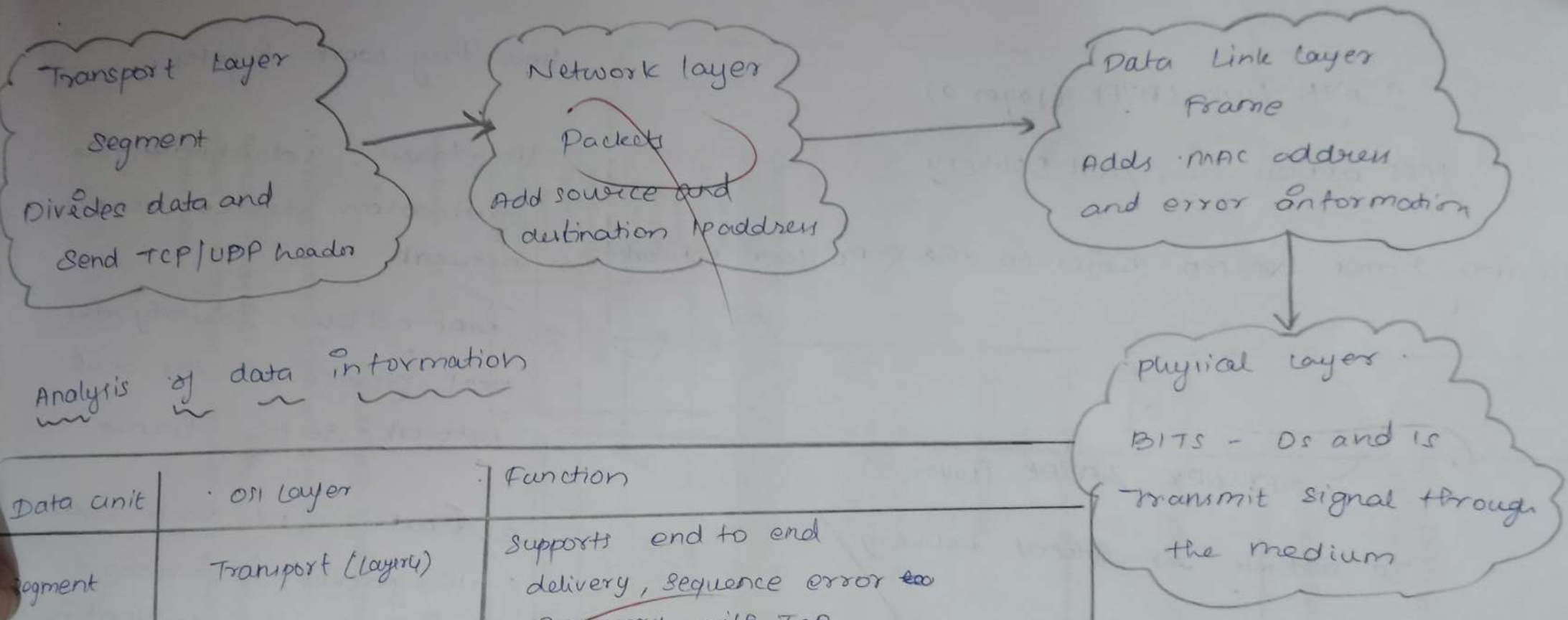
Session Layer: (Layer 5)

- Establishes, maintains, and terminate communication sessions
- control the dialogue between two devices
- Helps synchronize communication and resume

Data Link Layer (Layer 2):

- converts data into frames of transmision
- uses MAC addresses to deliver frames within local networks
- Detects transmision errors and manage access to the transmision medium.

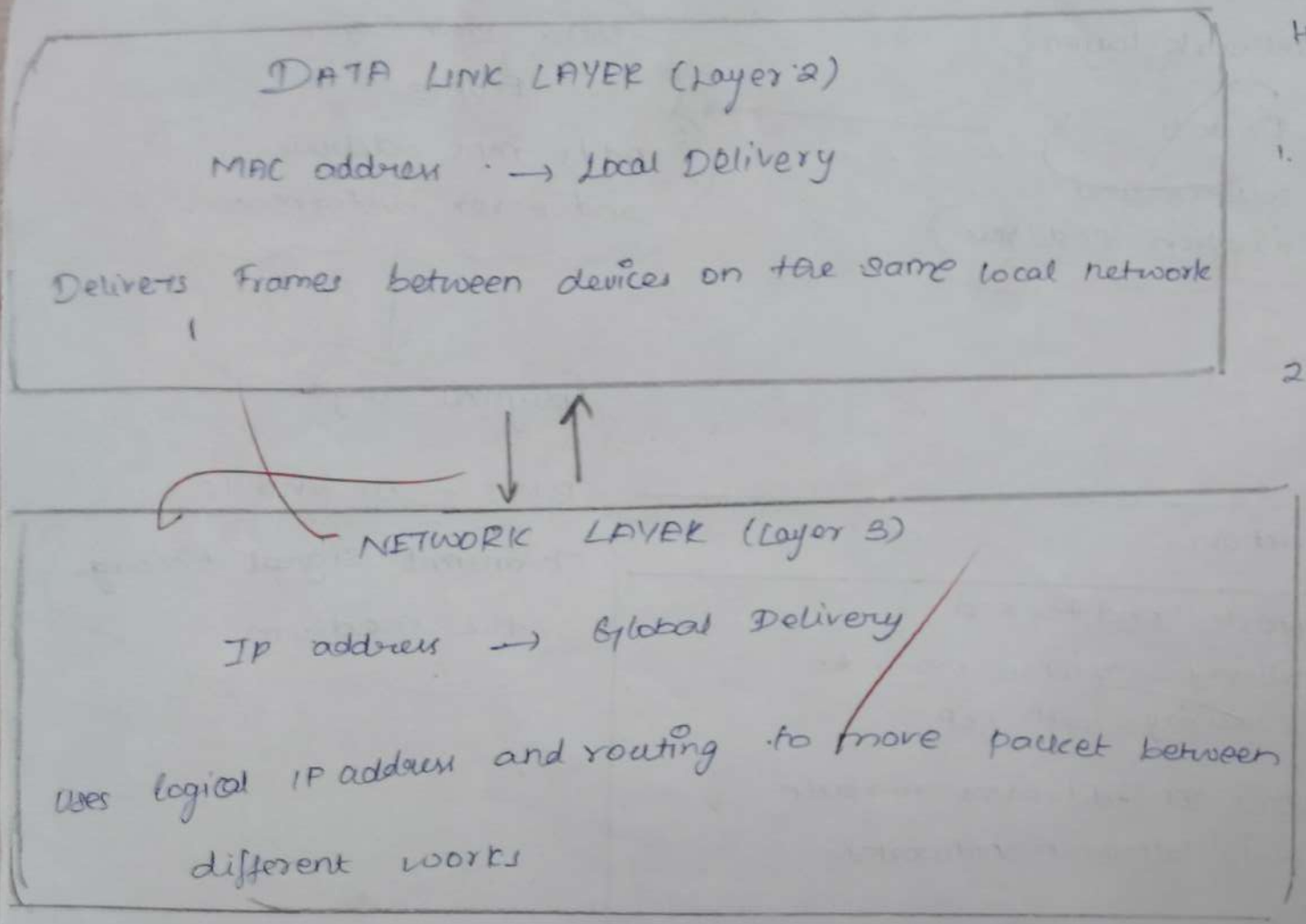
(8) Concept Mapping : Data unit and OSI Layer : Data transformation MAP



Analysis of data information

Data unit	OSI Layer	Function
Segment	Transport (Layer 4)	Supports end to end delivery, sequence error recovery with TCP
Packet	Network (Layer 3)	Uses IP addresses to route data between networks
Frame	Data link (Layer 2)	Delivers Data over local link and detects transmission error
Bits	physical (Layer 1)	Represent data as 0s and 1s transmitted as physical signal

Explanation : MAC address and IP address : The Data link layer and Network



How they work together:

1. IP address : Identifies the destination devices across network .
2. MAC address: Identify the next device on the local network so the frame
3. Routing : At each router, the packet is forward towards its destination, usually inside a new frame with updated mac address.